

Multimodal Computational Methods in Political Science

Images and Convolutional Neural Networks

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Why go beyond computational text analysis as a political scientist?

Introduction to Images

Recap: Neural Networks

Convolutional Neural Networks (CNN)

The Issue with Deeper CNNs and ResNet

Region-based CNNs for Object Detection

Image Preprocessing and CNN Autotaggers

Key Takeaways and What's Next

Why go beyond computational text analysis?

- increased use of images with digital space - many mostly used social media sites nowadays rely heavily on image, audio and video - including memes and AI-images

As Webb, Casas and Wilkerson (2020) summarizes previous research -including images in social science analysis is necessary because:

- Images capture attention more effectively than written or spoken content
- Visuals impact agenda-setting, political participation, perception of politicians
- They facilitate faster information processing and improve memory recall
- Visuals are uniquely capable of evoking strong emotional responses

From Text to Images: Same Logic, Different Data

Concept	Text Analysis	Image Analysis
<i>Basic Unit</i>	Token (Word/Subword)	Pixel
<i>Input Structure</i>	Sequence of Tokens	Grid of Pixels ($H \times W \times 3$)
<i>Representation</i>	Word Embedding	Feature Map
<i>Core Operation</i>	Attention over Tokens	Convolution over Patches
<i>Pretrained Model</i>	BERT, RoBERTa	ResNet, VGG
<i>Adaptation</i>	Fine-tune on labeled text	Fine-tune on labeled images

■ **Political text:** “*We stand together against the far right!*” → Tokenization → Sentiment / Frame Classification

■ **Political image:** Protest photo with banners → Pixel grid → Object Detection / Scene Classification

→ The core workflow is identical - but images capture what text cannot: bodies, symbols, scale, and emotion.

Introduction to Images (1/3)

- in computers images are represented by matrices or graphs displaying pixels
- file-types: JPEG, PNG, RAW, TIF, etc.
- quality of digitized image is dependent on sampling (number of pixels) and quantization (color depth)
- The largest and most often used labeled Image Dataset for CNN Pretraining is [ImageNet](#); [MS-Celeb-1M](#) contains celebrity labeled images; [COCO](#) contains images labeled by objects
- Additional Datasets can be found on [Kaggle](#) or [Huggingface](#) or [Harvard Dataverse](#)
- In social sciences we mostly use computer vision for: classification, object detection and sentiment analysis

Images can be characterized by:

- Spatial dimensions: 2D (traditional images) vs. 3D (volumetric data like satellite/drone imagery)
- Color system: RGB, grayscale, HSV, CMYK, or other color spaces
- Temporal component: Static (single frame/image) vs. dynamic (video/sequence of frames over time)

Recap: Neural Networks

You have already learned in Section 1 of this seminar, what a Neural Network is.

- Inspired by neurons in brains (artificial neurons are also called "perceptron")
- Neurons calculate weighted sum of input values
- A neural network learns by passing data forward through weighted layers (**forward pass**), computing the error via a **loss function**, and updating the weights backwards through the network (**backpropagation**) using **gradient descent**
- In a Dense Neural Networks - each layer of Neurons receives the same inputs → when analyzing images, Dense Neural Networks must flatten e.g. 2D image to 1D vector and each neuron gets weights for each input - meaning that when you have 1000 inputs and 10 neurons you will get 10 000 weights
- When more than one hidden layer → called "Deep Neural Network"

CNNs versus Dense NNs

Why CNNs for Image Analysis instead of NNs?

- For Dense NNs each input neuron multiplies with a different weight for each output neuron
- For Convolutional NNs the same kernel multiplies with many different patches of neurons, at different locations → local structure matters

- **Tidbit:** In most CNN libraries (e.g. PyTorch) we actually use Cross-Correlation instead of Convolution (Convolution would require the kernel to be flipped - since mathematically the order of the kernel should not change the result for convolution)

Convolutional Neural Networks (CNN)

CNNs rely on three core principles:

- Translation Invariance: Early layers recognize the same pattern anywhere in the image.
- Locality: They focus on small, local areas first.
- Hierarchy: Deeper layers aggregate these local details to understand the whole image. → Receptive Field refers to the fields neurons in each layer can see - meaning the deeper the CNN, the larger the receptive field for each added neuron layer
→ this leads to the intuition, that the deeper the CNN, the better it can represent complex features

Images as 3rd-order Tensors

- CNNs process data as 3rd-order tensors ($Height \times Width \times Channels$) - not flat images.
- The **Channel** dimension means different things at different stages:
 - **Input:** 3 fixed channels: Red, Green, Blue (RGB).
 - **Hidden Layers:** n learned channels, each encoding a detected feature (e.g., edges, textures, shapes).
- In other words: depth evolves from **raw color** $\rightarrow$ **abstract features** as data passes through the network.

Padding, Stride and Pooling

Padding

- Adds extra pixels (zeros) around input borders
- Preserves spatial dimensions of output
- Prevents edge information loss
- Enables deeper network architectures

Stride

- Steps a kernel shifts across the input per move
- Stride $S=1$: one element at a time
- Stride $S>1$: skips S elements per step
- Used for downsampling & efficiency
- Learned during training

Pooling

- Reduces spatial dimensions & locality sensitivity
- **Max pooling**: takes maximum in filter window (common)
- **Avg pooling**: takes arithmetic mean
- Custom stride & padding also applicable

The Issue with Deeper CNNs

Deeper models initially improved performance (AlexNet, VGG), but eventually depth became a liability: training error increased and accuracy saturated, suggesting a fundamental challenge with very deep networks.

The Problem

- Neural networks should improve when we add more layers, but empirically they often degrade
- Deeper networks don't simply extend what shallower networks learned - they represent fundamentally different function classes
- A deeper architecture may learn patterns that move away from the target function rather than toward it
- No guarantee that larger model capacity translates to better solutions

Why This Happens

- Expanding from a shallow to a deep network changes what the model can express
- The optimization landscape becomes more complex; the network can get stuck in worse local optima
- The larger function class doesn't necessarily contain the solutions found by the smaller class

The ResNet Solution: Skip Connections

- Skip connections: direct pathways that allow information to bypass intermediate layers
- Each residual block learns: $\text{output} = \text{input} + f(\text{input})$, where f is the new layers
- If f learns to be zero (do nothing), the output is just the input; if f learns something useful, it gets added on top
- New layers can either contribute useful transformations or simply pass information through (learning the identity mapping)
- Result: adding layers is monotonically non-decreasing in performance-depth becomes a free parameter to scale safely

- Pretrained CNNs (or others), that will generate image labels - possible bias issues
- Commercial (e.g. Amazon Rekognition, Microsoft Computer Vision, Google Cloud Vision) vs open-source (e.g. YOLOv8 (object detection), [face_recognition](#))
- While open-source Autotaggers might provide fewer labels and be trained for fewer tasks, but they can be adapted via e.g. fine-tuning - which we will discuss next week
- For commercial Autotaggers we do not know the existing labels until trying and we do not know the exact model and updates they use (issue of reproducibility)
- **But** many labels we are interested in are not currently covered by autotaggers **and** might not be accurate for our tasks

Region-based CNNs for Object Detection

- **R-CNN:** Propose regions, extract features per region, classify each with a CNN
- **Fast R-CNN:** Extracts features once from the full image; trains classification and localization jointly in a single end-to-end model
- **Faster R-CNN:** Introduces a Region Proposal Network (RPN), a convolutional mechanism that selectively proposes regions of interest, combined with Fast R-CNN
- **YOLO:** Divides the image into a grid and predicts bounding boxes and class labels for each cell simultaneously in a single forward pass

Object Detection: Two-Stage vs. Single-Stage

Two-Stage (e.g., Faster R-CNN)

- *Process:* Region Proposal → Classification
- **Pros:** High accuracy, precise localization
- **Cons:** Slower due to multi-stage pipeline

Single-Stage (e.g., YOLO)

- *Process:* Direct regression in one pass
- **Pros:** Extremely fast (Real-time)
- **Cons:** Challenging with small/dense objects

In our practical session we will use YOLOv8.

→ keeps the single-pass philosophy of the original YOLO but uses a much deeper, more sophisticated convolutional backbone and detects objects at multiple scales simultaneously → far more accurate on complex real-world images

Copyright (EU)

- No general fair use principle in EU law.
- DSM Directive allows data mining exemption for research - but under strict conditions

Privacy

- Images often contain personal data (faces, locations)
- **GDPR:** Legal basis required for collection & processing; anonymization must be considered

Harm & Algorithmic Bias

- Non-representative training data reproduces societal biases
- We must identify bias and anticipate harm before deployment

- Make images the same size (padding, cropping, squishing, rescaling)
- Normalization (scaling and centering) color intensity
- If beneficial do data augmentation - can decrease necessary training set size (e.g. random crop, horizontal flip)

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Thanks for your attention!